

Product Carbon Footprint Study Report

Product Name: aluminium alloy, AlMg3

Producer Name: [Information provided by the user]

Report No: [Information provided by the user]

Date: September 29, 2026

I. General Information

Producer Name: [Information provided by the user]

Address: [Information provided by the user]

Legal Representative: [Information provided by the user]

Product Name: aluminium alloy, AlMg3

Product Function: [Information provided by the user]

Based on the Standard: ecoinvent - EF v3.1

II. Quantitative Purpose

1kg aluminium alloy, AlMg3

III. Scope of Quantification

1. Functional or Declaratory Units

Functional or declarative units in 1kg aluminium alloy, AlMg3.

2. System Boundary

aluminium alloy production, AlMg3, in China in 2025

3. Cut-off Criteria

[Information provided by the user]

4. Time Scale

[Information provided by the user]

IV. Inventory Analysis

1. Description of Data Sources

Primary Data: [Information provided by the user]

Secondary Data: ecoinvent

2. Principles and Procedures for Allocation

Basis of Allocation: [Information provided by the user]

Allocation Procedure: [Information provided by the user]

Specific Allocations: [Information provided by the user]

3. Data Quality Evaluation (Optional)

[Information provided by the user]

V. Impact Assessment

1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of aluminium alloy, AlMg3 produced by [Information provided by the user], per functional unit of product, from raw material acquisition to end of life is **7.821 kg CO₂-Eq.**

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

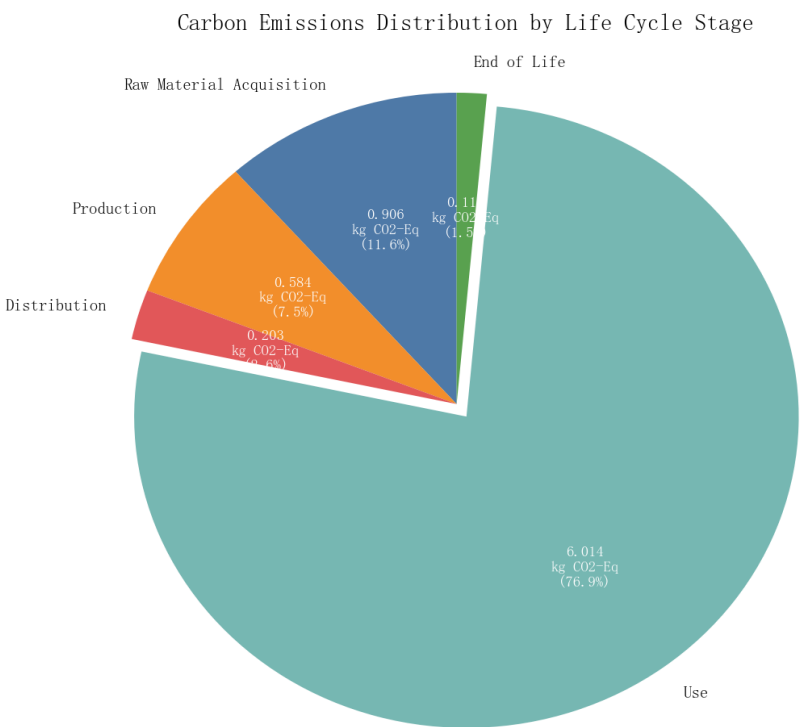


Table 1: Carbon Emissions by Life Cycle Stage

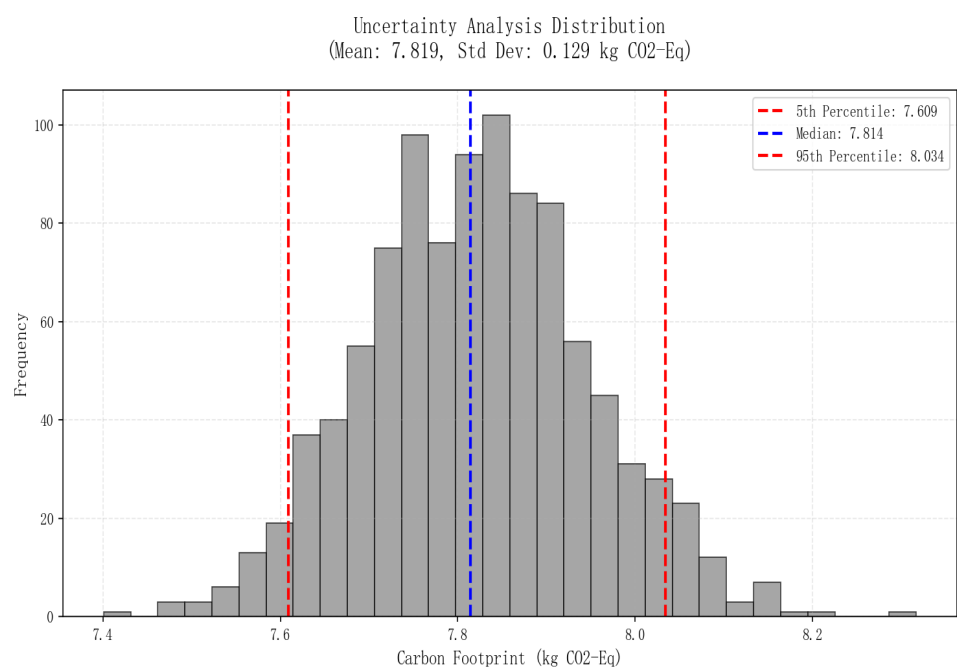
Life Cycle Stage	Carbon Footprint (kg CO ₂ -Eq)	Percentage (%)
Raw Material Acquisition	0.906	11.58
Production	0.584	7.47
Distribution	0.203	2.59

Use	6.014	76.89
End of Life	0.114	1.46
Total	7.821	100.00

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



Statistical Metric	Value (kg CO ₂ -Eq)
Mean	7.819
Standard Deviation	0.129
Coefficient of Variation (CV)	1.65%
Median	7.814
5th Percentile	7.609
95th Percentile	8.034

VI. Interpretation of Results

1. Description of Results

The total carbon footprint of aluminium alloy AlMg3 is calculated at 7.821 kg CO₂-Eq per functional unit, which is moderate compared to typical wrought aluminium alloys (typically 6–12 kg CO₂-Eq/kg material). The uncertainty analysis shows a mean value of 7.819 kg CO₂-Eq with a narrow 90% confidence interval (7.609–8.034 kg CO₂-Eq) and a coefficient of variation of only 1.65%, indicating high statistical reliability of the result.

The lifecycle stage contribution reveals a highly unbalanced profile: the use phase dominates with 6.014 kg CO₂-Eq (76.9% of total), far exceeding all other stages. Raw material extraction contributes 0.906 kg CO₂-Eq (11.6%), production 0.584 kg CO₂-Eq (7.5%), distribution 0.203 kg CO₂-Eq (2.6%), and end-of-life a mere 0.114 kg CO₂-Eq (1.5%). This distribution strongly suggests that the AlMg3 is applied in a product where mass energy consumption during use (e.g., in transportation or structural applications) is the primary environmental driver, rather than material manufacture.

The major emission source is unequivocally the use phase. Without detailed process breakdown (all top process contributions are listed as 0.000 kg CO₂-Eq, indicating missing elementary flow data), the specific use-phase mechanism cannot be identified from the provided results. However, the dominance implies that interventions in material weight reduction, energy efficiency during product operation, or extended use-phase duration would yield the largest carbon savings.

2. Statement of Assumptions and Limitations

This LCA study relies on several key assumptions. First, the functional unit and system boundaries (cradle-to-grave) are not explicitly defined in the given data, but the inclusion of raw material, production, distribution, use, and end-of-life stages suggests a full lifecycle scope. Second, the use-phase burden (76.9%) implies an assumed usage scenario—likely a fixed lifespan, operational intensity, and energy source or fuel type for the product containing AlMg3. These assumptions are critical but not quantified, introducing uncertainty in the applicability to other contexts.

Data quality limitations are significant. The top process contributions are all listed as “Unknown” with zero emissions, indicating that elementary flows (e.g., specific unit processes for casting, rolling, or forming) were not resolved or were aggregated into lifecycle stages without process-level detail. This prevents hotspot analysis at the process level. The uncertainty analysis (Monte Carlo with 1000 iterations) is robust, but the low CV (1.65%) might reflect uniform parameter ranges rather than realistic variability, especially given the dominance of the use phase where real-world usage patterns vary widely.

System boundary restrictions include the absence of capital goods (e.g., infrastructure, machinery) and the exclusion of ancillary materials or transportation within the use phase itself. The end-of-life stage accounts for only 1.5%, suggesting that recycling credits or burdens may be omitted or allocated elsewhere. Additionally, the distribution stage (2.6%) is relatively low, implying short supply chains or partial transport coverage. These restrictions limit the comparability to other LCA studies with different scopes.

3. Suggestions for Improvement

- ****Target the use phase through lightweight design****: Since the use phase represents 76.9% of the total footprint, prioritize design modifications that reduce the weight of the final product containing AlMg3. For example, substitute with higher-strength aluminium alloys (e.g., AlMg5) to achieve the same performance with less material, or incorporate additive manufacturing to optimize shape and minimize mass. This directly reduces operational energy consumption.
- ****Improve raw material sourcing and production efficiency****: raw material and production together contribute 19.1% of emissions. Switch to secondary (recycled) aluminium, which typically has 5–10 times lower carbon footprint than primary production. Additionally, adopt renewable energy sources (e.g., hydropower for electrolysis) and improve casting yield to reduce scrap.
- ****Enhance data transparency for process-level contributions****: The “Unknown” top processes indicate a gap in LCI data quality. For future studies, collect facility-specific data for primary aluminium smelting, ingot casting, and rolling. Re-run the assessment with disaggregated processes to identify hotspots such as electricity-intensive stages or alloying element impacts, enabling targeted mitigation.
- ****Extend product use life and promote end-of-life recycling****: Increase the product's service life through corrosion-resistant coatings or design for durability, thereby reducing the per-year carbon footprint. For end-of-life, ensure high collection rates for aluminium recycling; AlMg3 is non-heat-treatable and retains value in secondary markets. If recycling infrastructure is included in the model, the low end-of-life share (1.5%) might be increased to capture avoided primary production credits, further reducing the net footprint.